ORION SPORTS OS / ENGINEERING REPORT

FORMULA ONE PREDICTION SYSTEM

How the Python model works, where it lives, and how to change it

WORKED MODEL OUTPUT

FASTEST LAP	FASTEST-LAP CHANCE	PIT WINDOW	CONFIDENCE
1:28.521	69.9%	L26–31	29%

VERSION 1.0 · 22 AUGUST 2026

Python 3 standard library · Next.js API bridge · auditable CSV evidence

System architecture

ONE ENGINE, EXPLICIT BOUNDARIES

The interface never calculates a Formula One forecast in JavaScript. It asks the server to execute the Python model, parses its JSON response, and renders the result only inside the Prediction tab.

SCENARIO UI	→	NEXT.JS ROUTE	→	PYTHON ENGINE	→	JSON OUTPUT
driver · race · weather		validated query inputs		historical model		charts · strategy

SEPARATION OF CONCERNS

- Recorded facts: Jolpica results, standings, pit stops and lap timing feed Latest Race and Race Analysis.
- Forecast evidence: local CSV history feeds only f1_predictor.py.
- Presentation: React, Recharts, Framer Motion and React Three Fiber visualize outputs without altering them.

REQUEST PATH

```
GET /api/f1-predict?driver=VER&race=Abu%20Dhabi&trackTemp=34&rain=10&fuel=45&safetyCar=0
```

File map

WHERE EVERYTHING IS

MODEL	`DataBase/PredictionSystem/f1_predictor.py`	Owns every forecast calculation.
EXAMPLE	`DataBase/PredictionSystem/sample_prediction.json`	A reproducible output for learning and testing.
SERVER BRIDGE	`src/lib/f1PythonPrediction.ts`	Safely executes Python without a shell.
API	`src/app/api/f1-predict/route.ts`	Validates browser controls and returns JSON.
LIVE FACTS	`src/lib/f1RaceData.ts`	Fetches and caches the latest completed race.
USER INTERFACE	`src/components/FormulaOneDashboard.tsx`	Separates facts, analysis, drivers and predictions.
3D REPLAY	`src/components/CircuitReplay3D.tsx`	Animates recorded lap-duration pace.
CACHE	`DataBase/cache/formula-one/last-race.json`	Offline fallback for the last successful fetch.
THIS REPORT	`output/pdf/Orion_F1_Prediction_System_Report.pdf`	The document you are reading.

Evidence and feature engineering

THE ROWS BEHIND THE FORECAST

DRIVER_RACE_SUMMARY.CSV

best lap, pace consistency, best finish, sample depth

PIT_STRATEGY.CSV

driver/team pit-lap history and service evidence

PIT_STOPS.CSV

observed stop durations used for service-time baseline

COMPOUND_PERFORMANCE.CSV

soft/medium/hard pace and average tyre life

STINT_ANALYSIS.CSV

stint lengths and compound sequencing

DRIVERS.CSV

driver code, identity, nationality and number

RUNTIME PREPARATION

1. Parse CSV with Python's csv.DictReader; malformed or empty numeric values are ignored.
2. Filter evidence by selected driver and circuit; fall back to broader driver history when sparse.
3. Compute robust centers (medians/means), consistency and sample-depth signals.
4. Apply scenario penalties only after the historical baseline has been established.

PROVENANCE RULE

Live race facts are cached separately and never become claimed future facts. Model output carries confidence, scenario inputs and evidence counts so the UI can show what the estimate knows—and what it does not.

Model mathematics

TRANSPARENT BY DESIGN

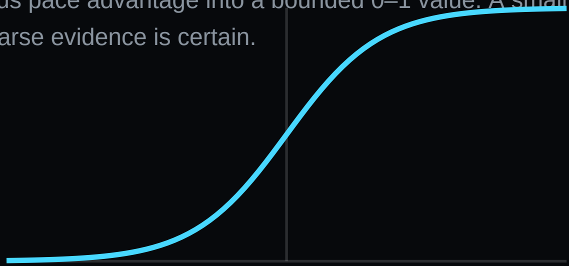
PACE ESTIMATE

```
lap = 0.68 × driver_best + 0.32 × circuit_benchmark
      + consistency_adjustment + temperature + rain + fuel
```

PROBABILITY CALIBRATION

The fastest-lap probability uses a sigmoid. That converts a continuous pace advantage into a bounded 0–1 value. A small sample penalty pulls confidence downward instead of pretending sparse evidence is certain.

```
p = 1 / (1 + exp(−score))
```



STRATEGY OUTPUTS

PIT WINDOW	Median historic pit lap, shifted by rain/safety-car scenario and clamped to a credible race range.
EXPECTED STOPS	Historical stop count plus scenario pressure; high rain can add a tyre transition.
TYRE PLAN	Compound performance and average stint life allocate sequential stints across the race.
PIT DURATION	Median observed service time for the selected evidence, with bounded fallback.
CONFIDENCE	Blend of driver and pit sample coverage, reduced for extrapolation and volatility.

Worked example

VER · ABU DHABI · DRY-BIASED SCENARIO

PREDICTED LAP	FASTEST-LAP CHANCE	PIT SERVICE	CONFIDENCE
1:28.521	69.9%	23.45s	29%

SCENARIO

Track temperature 34°C · rain 10% · fuel 45 kg · safety car off

TYRE TIMELINE

HARD · 13 laps	SOFT · 13 laps	MEDIUM · 16 laps

INTERPRETATION

- The modeled pit window opens on lap 26 and closes on lap 31.
- The engine recommends 2 stop(s); this changes when scenario controls imply transition tyres or safety-car timing.
- Re-running the same command with the same CSV files and inputs produces the same output.
- This output is a baseline estimate—not knowledge of future team strategy, incidents, upgrades or weather.

Run, modify and test

A PRACTICAL OPERATOR GUIDE

RUN PYTHON

```
python3 DataBase/PredictionSystem/f1_predictor.py --driver VER --race 'Abu Dhabi' --track-temp 34 --rain 10 --fuel 45
```

SAVE JSON

```
add: --output DataBase/PredictionSystem/sample_prediction.json
```

CALL THE API

```
curl 'http://localhost:3000/api/f1-predict?driver=VER&race=Abu%20Dhabi&trackTemp=34&rain=10&fuel=45&safetyCar=0'
```

VERIFY APP

```
npm run lint · npx tsc --noEmit · npm run build
```

SAFELY TUNE THE MODEL

- Edit weights and scenario coefficients in f1_predictor.py; change one family of coefficients at a time.
- Keep a fixed validation set of historical races and compare absolute lap error, pit-window error and probability calibration.
- Never train and evaluate on the same race rows. Prefer time-ordered validation to avoid future-data leakage.
- Regenerate sample_prediction.json and this report after a material model change.

Limits, sources and next steps

WHAT THE SYSTEM DOES NOT CLAIM

KNOWN LIMITS

- Historical CSV coverage may not represent the current car, regulation package or tyre construction.
- The free timing feed has lap timing but no centimeter-level GPS trace; the 3D circuit is therefore an illustrative reconstruction.
- Pit duration feeds can include anomalous long stops. The UI chart excludes durations above 60 seconds; the model uses bounded robust statistics.
- Driver images depend on Wikipedia/Wikimedia availability and may fall back to initials.
- Probability is calibration, not certainty. Outcomes depend on weather, traffic, incidents and team decisions.

PRIMARY REFERENCES

Jolpica F1 API documentation	https://github.com/jolpica/jolpica-f1/blob/main/docs/README.md
Jolpica race endpoints	https://github.com/jolpica/jolpica-f1/blob/main/docs/endpoints/races.md
OpenF1 API and telemetry concepts	https://openf1.org/
Python standard library	https://docs.python.org/3/library/

RECOMMENDED NEXT ITERATION

Backtest each season with time-ordered folds, record calibration curves, add circuit geometry from a licensed coordinate source, and version every trained parameter set. Keep the current transparent baseline as the benchmark that more complex models must beat.